

Documentation (Roman Urdu)

Yeh file poorey RAG (Retrieval-Augmented Generation) system ka "Master Orchestrator / Project Manager" aei!

File Ka Maqsad (Purpose)

Yeh baaki saari tools (Chunking, Embedding, Retriever, Ingest) ko aik saath bula ker aik mukkamal step-by-step pipeline banatti ha, taka baahir k system (jasy UI) sirf 1 simple function command karen ar sb kam automatic hz jyain.

Code Blocks aur Functions ki Tafseel

1. ingest_document()

Tafseel: Jab nayia doc upload hoa tu:

1. Pehle `ingest_file` py bhejti hs parse huny.
2. Pher `chunking_engine` sa tukre karti e.
3. Phr Batch-by-Batch unky `embeddings_engine` mn float vectors niklti k.
4. Uske baadd faiss `vector_store` mn arr `keyword_index` (BM25) douono men save kardati ah!

2. query()

Tafseel: Jab user sawal pochta a tu:

1. Pehlai `hybrid_retriever` ko awaz dttui jo vector arr db sy answers match uthttaa u.
2. Us k resultys ko `reranker` sa filter krety ta ke fuzool chunks ht jayen.
3. Akhir me `citation_tracker` say Hawalajat (Sources/References [1], [2]) mark krke prompt k liy context ready kardi hz!

3. distilled_query()

- Aglai level ki Search (Distillation k satha) call kari ah hy jo complex questions toor drri u.